

Clinical Risk Assessment Report: Patient 51

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 32.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.48x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the 75th Percentile Tumor Density (CONVENTIONAL_HUQ3), which elevated the patient's absolute risk by +5.2%. Biologically, this feature reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

2. Key Pathological & Imaging Drivers (Elevating Risk)

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.88 [Top 16%] (+5.2%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.53 [Avg (68th %ile)] (+4.1%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.51 [Avg (75th %ile)] (+2.9%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.67 [Top 24%] (+2.8%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.05 [Top 14%] (+2.5%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 71.0 [Top 22%] (-2.0%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

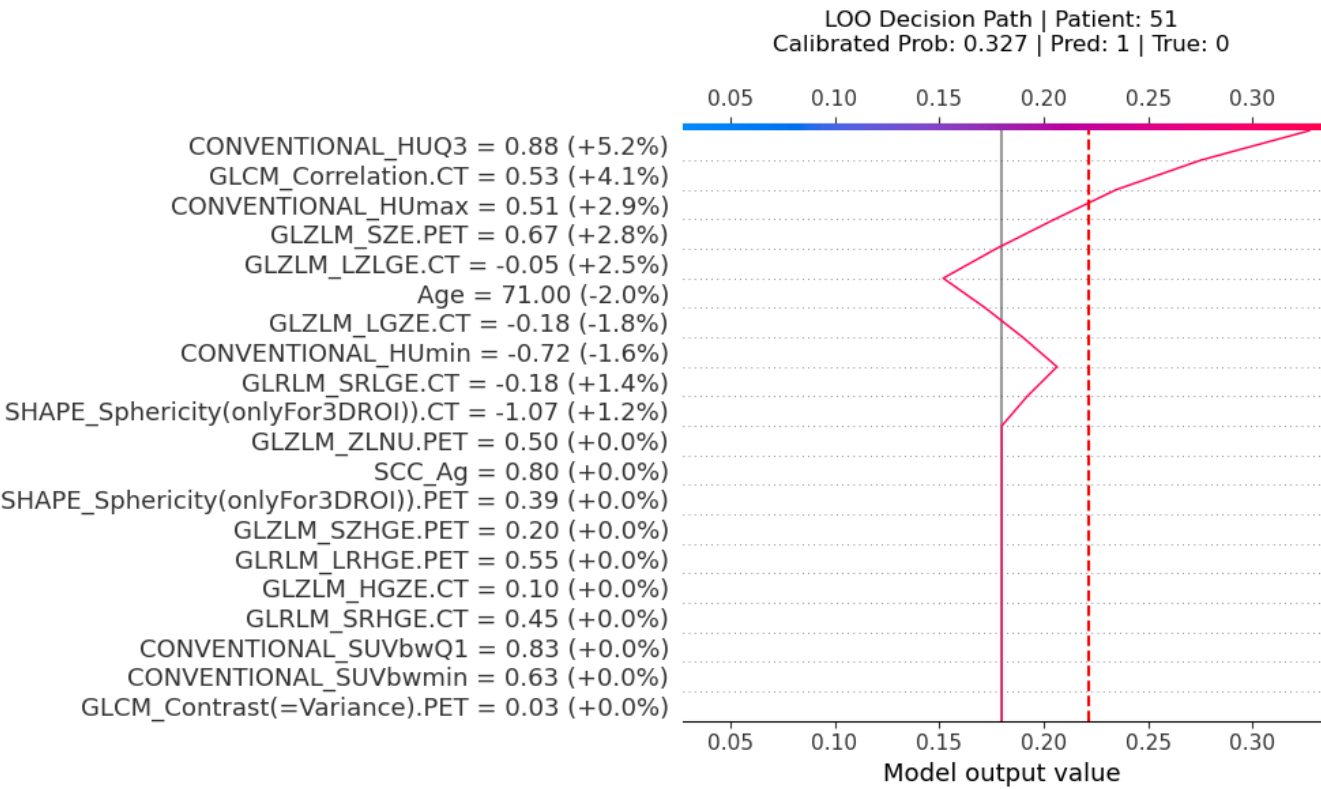
Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.18 [Avg (54th %ile)] (-1.8%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.72 [Bottom 23%] (-1.6%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

